where the next-to-last map is induced by the quotient map $X^n \to J_n(X)$. It is clear that φ is a ring homomorphism since the product in $J(X)$ is induced from the natural map $X^m \times X^n \to X^{m+n}$. To show that φ is an isomorphism, consider the following commutative diagram of short exact sequences:

$$\begin{tikzcd}
0 \arrow[r] & T_{n-1}\widetilde{H}_*(X) \arrow[r] \arrow[d, "\varphi"'] & T_n\widetilde{H}_*(X) \arrow[r] \arrow[d, "\varphi"'] & \widetilde{H}_*(X)^{\otimes n} \arrow[r] \arrow[d, "\times"', "\approx"] & 0 \\
0 \arrow[r] & H_*(J_{n-1}(X)) \arrow[r] & H_*(J_n(X)) \arrow[r] & \widetilde{H}_*(X^{\wedge n}) \arrow[r] & 0
\end{tikzcd}$$

In the upper row, $T_m\widetilde{H}_*(X)$ denotes the direct sum of the products $\widetilde{H}_*(X)^{\otimes k}$ for $k \le m$, so this row is exact. The second row is the homology exact sequence for the pair $(J_n(X), J_{n-1}(X))$, with quotient $J_n(X)/J_{n-1}(X)$ the n-fold smash product $X^{\wedge n}$. This long exact sequence breaks up into short exact sequences as indicated, by commutativity of the right-hand square and the fact that the right-hand vertical map is an isomorphism by the Künneth formula, using the hypothesis that $H_*(X)$ is free over the given coefficient ring. By induction on n and the five-lemma we deduce from the diagram that $\varphi : T_n\widetilde{H}_*(X) \to H_*(J_n(X))$ is an isomorphism for all n. Letting n go to ∞, this implies that $\varphi : T\widetilde{H}_*(X) \to H_*(J(X))$ is an isomorphism since in any given dimension $T_n\widetilde{H}_*(X)$ is independent of n when n is sufficiently large, and the same is true of $H_*(J_n(X))$ by the second row of the diagram. ◻

Dual Hopf Algebras

There is a close connection between the Pontryagin product in homology and the Hopf algebra structure on cohomology. Suppose that X is an H–space such that, with coefficients in a field R, the vector spaces $H_n(X;R)$ are finite-dimensional for all n. Alternatively, we could take $R = \mathbb{Z}$ and assume $H_n(X;\mathbb{Z})$ is finitely generated and free for all n. In either case we have $H^n(X;R) = \mathrm{Hom}_R(H_n(X;R), R)$, and as a consequence the Pontryagin product $H_*(X;R) \otimes H_*(X;R) \to H_*(X;R)$ and the coproduct $\Delta : H^*(X;R) \to H^*(X;R) \otimes H^*(X;R)$ are dual to each other, both being induced by the H–space product $\mu : X \times X \to X$. Therefore the coproduct in cohomology determines the Pontryagin product in homology, and vice versa. Specifically, the component $\Delta_{ij} : H^{i+j}(X;R) \to H^i(X;R) \otimes H^j(X;R)$ of Δ is dual to the product $H_i(X;R) \otimes H_j(X;R) \to H_{i+j}(X;R)$.

Example 3C.9. Consider $J(S^n)$ with n even, so $H^*(J(S^n);\mathbb{Z})$ is the divided polynomial algebra $\Gamma_{\mathbb{Z}}[\alpha]$. In Example 3C.5 we derived the coproduct formula $\Delta(\alpha_k) = \sum_i \alpha_i \otimes \alpha_{k-i}$. Thus Δ_{ij} takes α_{i+j} to $\alpha_i \otimes \alpha_j$, so if x_i is the generator of $H_{in}(J(S^n);\mathbb{Z})$ dual to α_i, then $x_i x_j = x_{i+j}$. This says that $H_*(J(S^n);\mathbb{Z})$ is the polynomial ring $\mathbb{Z}[x]$. We showed this in Example 3C.7 using the cell structure of $J(S^n)$, but the present proof deduces it purely algebraically from the cup product structure.

Now we wish to show that the relation between $H^*(X;R)$ and $H_*(X;R)$ is perfectly symmetric: They are *dual Hopf algebras*. This is a purely algebraic fact:

Proposition 3C.10. *Let A be a Hopf algebra over R that is a finitely generated free R-module in each dimension. Then the product $\pi: A\otimes A \to A$ and coproduct $\Delta: A \to A\otimes A$ have duals $\pi^*: A^* \to A^*\otimes A^*$ and $\Delta^*: A^*\otimes A^* \to A^*$ that give A^* the structure of a Hopf algebra.*

Proof: This will be apparent if we reinterpret the Hopf algebra structure on A formally as a pair of graded R-module homomorphisms $\pi: A\otimes A \to A$ and $\Delta: A \to A\otimes A$ together with an element $1 \in A^0$ satisfying:

(1) The two compositions $A \xrightarrow{i_\ell} A\otimes A \xrightarrow{\pi} A$ and $A \xrightarrow{i_r} A\otimes A \xrightarrow{\pi} A$ are the identity, where $i_\ell(a) = a\otimes 1$ and $i_r(a) = 1\otimes a$. This says that 1 is a two-sided identity for the multiplication in A.

(2) The two compositions $A \xrightarrow{\Delta} A\otimes A \xrightarrow{p_\ell} A$ and $A \xrightarrow{\Delta} A\otimes A \xrightarrow{p_r} A$ are the identity, where $p_\ell(a\otimes 1) = a$, $p_\ell(a\otimes b) = 0$ if $b \in A^j$ with $j > 0$, $p_r(1\otimes a) = a$, and $p_r(a\otimes b) = 0$ if $a \in A^j$ with $j > 0$. This is just the coproduct formula $\Delta(a) = a\otimes 1 + 1\otimes a + \sum_{0<i<n} a'_i \otimes a''_{n-i}$.

(3) The diagram at the right commutes, where $\tau(a\otimes b\otimes c\otimes d) = (-1)^{ij} a\otimes c\otimes b\otimes d$ for $b \in A^i$, $c \in A^j$. This is the condition that Δ is an algebra homomorphism since if we follow an element $a\otimes b \in A^m\otimes A^n$ across the top of the diagram we get $\Delta(ab)$, while the lower route gives first $\Delta(a)\otimes\Delta(b) = \big(\sum_i a'_i\otimes a''_{m-i}\big)\otimes\big(\sum_j b'_j\otimes b''_{n-j}\big)$, then after applying τ and $\pi\otimes\pi$ this becomes $\sum_{i,j}(-1)^{(m-i)j} a'_i b'_j \otimes a''_{m-i} b''_{n-j} = \big(\sum_i a'_i\otimes a''_{m-i}\big)\big(\sum_j b'_j\otimes b''_{n-j}\big)$, which is $\Delta(a)\Delta(b)$.

$$\begin{tikzcd} A\otimes A \arrow[r, "\pi"] \arrow[d, "\Delta\otimes\Delta"] & A \arrow[r, "\Delta"] & A\otimes A \\ A\otimes A\otimes A\otimes A \arrow[rr, "\tau"] & & A\otimes A\otimes A\otimes A \arrow[u, "\pi\otimes\pi"] \end{tikzcd}$$

Condition (1) for A dualizes to (2) for A^*, and similarly (2) for A dualizes to (1) for A^*. Condition (3) for A dualizes to (3) for A^*. $\square$

Example 3C.11. Let us compute the dual of a polynomial algebra $R[x]$. Suppose first that x has even dimension. Then $\Delta(x^n) = (x\otimes 1 + 1\otimes x)^n = \sum_i \binom{n}{i} x^i\otimes x^{n-i}$, so if α_i is dual to x^i, the term $\binom{n}{i}x^i\otimes x^{n-i}$ in $\Delta(x^n)$ gives the product relation $\alpha_i\alpha_{n-i} = \binom{n}{i}\alpha_n$. This is the rule for multiplication in a divided polynomial algebra, so the dual of $R[x]$ is $\Gamma_R[\alpha]$ if the dimension of x is even. This also holds if $2 = 0$ in R, since the even-dimensionality of x was used only to deduce that $R[x]\otimes R[x]$ is strictly commutative.

In case x is odd-dimensional, then as we saw in Example 3C.1, if we set $y = x^2$, we have $\Delta(y^n) = (y\otimes 1 + 1\otimes y)^n = \sum_i \binom{n}{i} y^i\otimes y^{n-i}$ and $\Delta(xy^n) = \Delta(x)\Delta(y^n) = \sum_i \binom{n}{i} xy^i\otimes y^{n-i} + \sum_i \binom{n}{i} y^i\otimes xy^{n-i}$. These formulas for Δ say that the dual of $R[x]$ is $\Lambda_R[\alpha]\otimes\Gamma_R[\beta]$ where α is dual to x and β is dual to y.

This algebra allows us to deduce the cup product structure on $H^*(J(S^n);R)$ from the geometric calculation $H_*(J(S^n);R) \approx R[x]$ in Example 3C.7. As another application, recall from earlier in this section that $\mathbb{R}\mathrm{P}^\infty$ and $\mathbb{C}\mathrm{P}^\infty$ are H–spaces, so from their cup product structures we can conclude that the Pontryagin rings $H_*(\mathbb{R}P^\infty;\mathbb{Z}_2)$ and $H_*(\mathbb{C}P^\infty;\mathbb{Z})$ are divided polynomial algebras.

In these examples the Hopf algebra is generated as an algebra by primitive elements, so the product determines the coproduct and hence the dual algebra. This is not true in general, however. For example, we have seen that the Hopf algebra $\Gamma_{\mathbb{Z}_p}[\alpha]$ is isomorphic as an algebra to $\bigotimes_{i\ge 0}\mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$, but if we regard the latter tensor product as the tensor product of the Hopf algebras $\mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$ then the elements α_{p^i} are primitive, though they are not primitive in $\Gamma_{\mathbb{Z}_p}[\alpha]$ for $i>0$. In fact, the Hopf algebra $\bigotimes_{i\ge 0}\mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$ is its own dual, according to one of the exercises below, but the dual of $\Gamma_{\mathbb{Z}_p}[\alpha]$ is $\mathbb{Z}_p[\alpha]$.

Exercises

1. Suppose that X is a CW complex with basepoint $e\in X$ a 0-cell. Show that X is an H–space if there is a map $\mu: X\times X\to X$ such that the maps $X\to X$, $x\mapsto\mu(x,e)$ and $x\mapsto\mu(e,x)$, are homotopic to the identity. [Sometimes this is taken as the definition of an H–space, rather than the more restrictive condition in the definition we have given.] With the same hypotheses, show also that μ can be homotoped so that e is a strict two-sided identity.

2. Show that a retract of an H–space is an H–space if it contains the identity element.

3. Show that in a homotopy-associative H–space whose set of path-components is a group with respect to the multiplication induced by the H–space structure, all the path-components must be homotopy equivalent. [Homotopy-associative means associative up to homotopy.]

4. Show that an H–space or topological group structure on a path-connected, locally path-connected space can be lifted to such a structure on its universal cover. [For the group $SO(n)$ considered in the next section, the universal cover for $n>2$ is a 2-sheeted cover, a group called $Spin(n)$.]

5. Show that if (X,e) is an H–space then $\pi_1(X,e)$ is abelian. [Compare the usual composition $f\cdot g$ of loops with the product $\mu(f(t),g(t))$ coming from the H–space multiplication μ.]

6. Show that S^n is an H–space iff the attaching map of the $2n$-cell of $J_2(S^n)$ is homotopically trivial.

7. What are the primitive elements of the Hopf algebra $\mathbb{Z}_p[x]$ for p prime?

8. Show that the tensor product of two Hopf algebras is a Hopf algebra.

9. Apply the theorems of Hopf and Borel to show that for an H–space X that is a connected finite CW complex with $\widetilde{H}_*(X;\mathbb{Z})\ne 0$, the Euler characteristic $\chi(X)$ is 0.

10. Let X be a path-connected H–space with $H^*(X;R)$ free and finitely generated in each dimension. For maps $f,g: X\to X$, the product $fg: X\to X$ is defined by $(fg)(x)=f(x)g(x)$, using the H–space product.